

Francesca Giardine

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EDUCATION

University of California, Berkeley

Doctor of Philosophy in Electrical Engineering and Computer Science

Advised by Dr. Robert Pilawa-Podgurski

Berkeley, CA

Aug. 2020 – May 2026

Smith College

Bachelor of Science in Engineering Science

Bachelor of Arts in Mathematics and Statistics

Summa Cum Laude

Northampton, MA

Aug. 2016 – May 2020

HONORS AND AWARDS

National Science Foundation Graduate Research Fellowship

Sept. 2022 - Sept. 2025

Hearts to Humanity Eternal Graduate Research Grant

Sept. 2023 - Sept. 2024

Berkeley EECS Department Fellowship

Aug. 2020

Best Paper Award, ECCE Europe 2024

September 2024

NextProf Nexus

August 2024

Adeline Devor Penberthy, Class of 1945, Memorial Prize

May 2020

Phi Beta Kappa Honor Society

May 2020

Tau Beta Kappa Engineering Honor Society

October 2018

Mu Sigma Rho Statistics Honor Society

May 2020

Sigma Xi Research Honor Society

May 2020

2019 Five College DataFest Best in Group

March 2019

PUBLICATIONS

Journal Publications

1. N. C. Brooks, **F. Giardine** and R. C. N. Pilawa-Podgurski, "DC-Link Capacitors for Twice-Line Frequency Power Decoupling: Design-Oriented Figures-of-Merit with Empirical Application," *IEEE Transactions on Power Electronics*, Volume: 39, pp. 6569 - 6573 Issue: 6, June 2024
2. J. Ford, M. Paretto, D. Kotys-Schwartz, S. Howe, C. Gewirtz, J. Deters, T. Mahmud Chowdhury, R. Ott, N. Alvarez, D. Knight, C. Hernandez, L. Rosenbauer, A. Kary, **F. Giardine**. "Transitioning from Capstone Design Courses to Workplaces: A Study of New Engineers' First Three Months." *International Journal of Engineering Education*, June 2019

Peer-Reviewed Conference Publications

1. **F. Giardine**, S. Krishnan, M. Nerenberg and R. C. N. Pilawa-Podgurski, "On the Scaling of Common-Mode EMI in the Flying Capacitor Multilevel Converter" *2025 IEEE Workshop on Control and Modeling of Power Electronics (COMPEL)* June. 2025 [Accepted]
2. E. Krause, N. Biesterfeld, **F. Giardine**, and R. C. N. Pilawa-Podgurski, "Analysis of Steady-State Balancing in the Flying Capacitor Multilevel Converter Considering Capacitor Voltage Ripple" *2025 IEEE Workshop on Control and Modeling of Power Electronics (COMPEL)* June. 2025 [Accepted]

3. S. Krishnan, **F. Giardine**, J. Zou, R. Abramson, and R. C. N. Pilawa-Podgurski, "Design and Implementation of a GaN-based, Cascaded Isolated DC-DC Converter for Satellite Applications" *2025 IEEE/AIAA Transportation Electrification Conference & Electric Aircraft Technologies Symposium* June. 2025 [To Appear]
4. **F. Giardine**, S. Krishnan, L. Horowitz, and R. C. N. Pilawa-Podgurski, "A Variable Frequency Technique for EMI and Efficiency Improvements in High-Level Count Flying Capacitor Multilevel Converters" *2025 IEEE Applied Power Electronics Conference*, Mar. 2025
5. **F. Giardine**, Y. Wu and R. C. N. Pilawa-Podgurski, "A Variable Switching Frequency Control Technique for DC-AC Flying Capacitor Multilevel Converters to Improve Efficiency and Inductor Utilization" *2024 IEEE Energy Conversion Congress and Exposition*, Oct. 2024
6. **F. Giardine**, K. Fernandez and R. C. N. Pilawa-Podgurski, "A Two-Stage Non-Isolated Hybrid Switched-Capacitor Microinverter Utilizing a Fixed-Ratio Resonant DC-DC Stage with Startup Functionality and Flying Capacitor Multilevel DC-AC Stage" *2024 IEEE Energy Conversion Congress and Exposition - Europe*, Sept. 2024 (**Best Paper Award**)
7. **F. Giardine**, N. C. Brooks, K. Fernandez and R. C. N. Pilawa-Podgurski, "Utilizing Harmonic Injection to Reduce Energy Storage and Required Capacitance in an Active Series-Stacked Energy Buffer for Single-Phase Systems," *2022 IEEE 23rd Workshop on Control and Modeling for Power Electronics (COMPEL)*, June 2022
8. C. Gerwitz, **F. Giardine**, R. Ott, A. Kary. "Women's Unique Challenges in the Transitions to Engineering Work." *2020 ASEE Annual Conference & Exposition*, June 2020
9. S. Howe, M. Parette, R. Ott, J. Deters, C. Gewirtz, **F. Giardine**, C. Hernandez, A. Kary. "Women's Experiences in the Transition from Capstone Design Courses to Engineering Workplaces." *2019 ASEE Annual Conference & Exposition*, June 2019

TEACHING

University of California, Berkeley EECS

EE 108: Introduction to Electric Power and Renewable Energy	<i>Spring 2024, Spring 2025</i>
Certificate in Teaching and Learning in Higher Education	<i>Expected, Fall 2025</i>

Smith College

Engineering Master Tutor	<i>Spring 2019 - Spring 2020</i>
General Quantitative Tutor	<i>Fall 2018</i>

RESEARCH EXPERIENCE

Pilawa Research Group

<i>University of California, Berkeley</i>	<i>Aug. 2020 - Present</i> <i>Berkeley, CA</i>
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- Developing advanced hybrid switched capacitor converters for use in DC-AC grid-connected systems.
- Validation of fundamental converter and control behavior in simulation and hardware.

Power Electronic Device Characterization Internship

<i>National Renewable Energy Laboratory</i>	<i>June 2023 - Aug. 2023</i> <i>Golden, CO</i>
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- Designed a test setup for switch characterization for power electronics.
- Performed simulation, printed circuit board layout, and investigated advanced sensing techniques to achieve high-bandwidth performance.

Researcher in the Renewable and Appropriate Energy Lab *July 2020 - Aug. 2020*
University of California, Berkeley *Berkeley, CA*

- Continued research from the SUPERB REU under Professor Daniel Kammen in the Energy Resources Group.
- Performed statistical and spatial analyses in Python and GIS to isolate candidate communities for microgrid development

SUPERB Research Experience for Undergraduates *June 2019 - Aug. 2019*
University of California, Berkeley *Berkeley, CA*

- Analyzed geospatial data in RStudio and GIS to classify community types in California with emphasis on environmental predictors.
- Selected as a representative to present research at NSF REU Research Symposium in October 2019.

Capstone to Work Research Team *Sept. 2018-May 2020*
Smith College *Northampton, MA*

- Reviewed qualitative and quantitative data from a large pool of first year engineering employees to analyze the impacts of the engineering capstone experiences on employment after college.

SERVICE AND LEADERSHIP

University of California, Berkeley

Berkeley EECS Graduate Admissions Student Reader *2023, 2024*
 IEEE Power and Energy Society *Jan. 2022 - Present*

- 2023 Chair
- 2022 Vice Chair

EECS Graduate Student Association Committee Member *Sept. 2021 - Sept. 2022*

Smith College

Tau Beta Kappa Engineering Honor Society (Co-President) *Sept. 2019 - May 2020*
 Writing and Public Discourse Committee Student Liaison *Sept. 2018 - May 2020*
 Outdoor Orientation Leader *Fall 2019*

MENTORING

Nikhil Ograin (Undergraduate) *September 2024 - Present*

A Resonant Flying Capacitor Multilevel Inverter

Yayun Zhao (Undergraduate) *October 2024 - Present*

Design of a High-Performance Totem-Pole PFC Circuit